Every Prime Except 2 and 5 Divides All Rotations of Some Champernowne Prefix

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Abstract

Let $\mathcal{C}(m)$ denote the decimal integer obtained by concatenating $1, 2, \ldots, m$, and let $\mathcal{L}(m)$ be its number of decimal digits. We prove that, for every prime $p \notin \{2, 5\}$, some prefix $\mathcal{C}(m)$ and every cyclic digit rotation of that prefix are divisible by p. This settles the positivity conjecture for OEIS A181373. Our proof is elementary and constructive: for $p \geq 7$ it supplies an $m \leq 10^{p-1} + p(p-1) - 1$. Moreover, infinitely many such m exist for each admissible prime. A second construction in terms of the multiplicative order of 10 gives a sometimes sharper bound.

1 Introduction

Write

$$\mathcal{C}(m) = \underbrace{123456789101112\cdots m}_{\text{concatenated decimal blocks}}, \qquad \mathcal{L}(m) = \sum_{k=1}^{m} \bigl(1 + \lfloor \log_{10} k \rfloor\bigr).$$

OEIS A007908 records $\mathcal{C}(m)$ and A058183 records $\mathcal{L}(m)$ [2, 3]. In A181373, the n-th term is the least $m \geq 1$ such that the n-th prime divides every cyclic *digit* rotation of $\mathcal{C}(m)$, or 0 if no such m exists [1]. Chai Wah Wu proposed in 2023 that the only zero terms are those corresponding to 2 and 5.

The circular concatenations and their modular divisibility patterns have been investigated by Ripà and Dalmasso [4] and later revisited by Ripà [5]. We give a direct existence proof by forcing the digit count into a prescribed congruence class and then sweeping every residue of a prime field inside a single fixed-digit-length block.

2 A rotation criterion and an affine shift

For a decimal integer N of length D, cut its first h digits into an integer A, leaving a suffix B of length $D - h$. Thus $N = A10^{D-h} + B$. Moving the first h digits to the end gives $R_h = B10^h + A$, whence

$$R_h = 10^h N - A(10^D - 1). \tag{1}$$

This formula remains valid if the rotated digit string begins with zero: its integer value is still R_h. Consequently,

$$p \mid N \quad \text{and} \quad p \mid (10^D - 1) \quad \Longrightarrow \quad p \mid R_h \quad \text{for every } h. \tag{2}$$

OEIS also records the equivalence between the full-rotation condition and simultaneous divisibility in (2) [1]; we only need sufficiency.

Lemma 1 (Constant-step shift)**.** *Let $p \geq 7$ be prime and let r be a positive integer such that $p \nmid r$ and $10^r \equiv 1 \pmod{p}$. Suppose $m, m+1, \ldots, m+r$ are all d-digit positive integers, where $d \equiv 1 \pmod{r}$. Then*

$$\mathcal{C}(m+r) - \mathcal{C}(m) \equiv -\frac{r}{9} \pmod{p}. \tag{3}$$

Proof. Throughout the fixed-digit-length interval, concatenation satisfies

$$\mathcal{C}(t+1) = 10^d \mathcal{C}(t) + (t+1).$$

Since $d \equiv 1 \pmod{r}$ and $10^r \equiv 1 \pmod{p}$, we have $10^d \equiv 10 \pmod{p}$. Iterating r steps gives

$$\mathcal{C}(m+r) \equiv 10^r \mathcal{C}(m) + \sum_{i=1}^{r} (m+i) 10^{r-i} \pmod{p}.$$

Set $A = \sum_{i=1}^{r} i 10^{r-i}$. As $10 \not\equiv 1 \pmod{p}$ and $10^r \equiv 1 \pmod{p}$,

$$\sum_{i=1}^{r} 10^{r-i} = \frac{10^r - 1}{9} \equiv 0 \pmod{p}.$$

A telescoping calculation yields

$$9A = \sum_{i=1}^{r} 10^i - r = 10\frac{10^r - 1}{9} - r \equiv -r \pmod{p}.$$

Here 9 is invertible modulo p. Removing the term proportional to m proves (3). □

3 Resolution of the A181373 conjecture

Theorem 1. *For every prime $p \notin \{2, 5\}$, there exist infinitely many positive integers m for which p divides $\mathcal{C}(m)$ and all of its cyclic digit rotations. No such m exists for $p = 2$ or $p = 5$. More quantitatively, for every prime $p \geq 7$, such an m can be chosen with*

$$m \leq 10^{p-1} + p(p-1) - 1 < 10^p. \tag{4}$$

Consequently, A181373$(n) > 0$ *exactly when $n \notin \{1, 3\}$.*

Proof. Fix a prime $p \geq 7$, and put

$$r = p - 1, \qquad d = p, \qquad L = 10^{p-1}, \qquad H = \mathcal{L}(L-1).$$

By Fermat's little theorem, $10^r \equiv 1 \pmod{p}$; furthermore $d \equiv 1 \pmod{r}$ and $p \nmid r$. Choose the unique $\delta \in \{0, 1, \ldots, r-1\}$ satisfying

$$\delta \equiv -H - 1 \pmod{r}, \tag{5}$$

and consider p endpoints

$$m_j = L + \delta + jr, \qquad j = 0, 1, \ldots, p-1. \tag{6}$$

All m_j have exactly $d = p$ decimal digits: indeed,

$$0 \leq \delta + jr \leq (r-1) + (p-1)r = p(p-1) - 1 < 9 \cdot 10^{p-1}.$$

(The last inequality is immediate for $p \geq 7$.) Consequently, by counting all d-digit blocks between L and m_j,

$$\mathcal{L}(m_j) = H + d(m_j - L + 1) = H + p(\delta + jr + 1) \equiv H + \delta + 1 \equiv 0 \pmod{r}.$$

It follows that

$$10^{\mathcal{L}(m_j)} \equiv 1 \pmod{p}. \tag{7}$$

Applying Lemma 1 successively inside this interval, using $r = p - 1$, shows

$$\mathcal{C}(m_j) \equiv \mathcal{C}(m_0) - \frac{jr}{9} \equiv \mathcal{C}(m_0) + \frac{j}{9} \pmod{p}.$$

As j varies from 0 to $p-1$, the right side takes every residue modulo p exactly once. In fact, choose the unique

$$j \equiv -9\mathcal{C}(m_0) \pmod{p}, \qquad 0 \leq j < p. \tag{8}$$

For this j, we obtain $p \mid \mathcal{C}(m_j)$, while (7) and (2) imply that p divides all digit rotations of $\mathcal{C}(m_j)$. The bound (4) follows from (6).

To obtain infinitely many distinct witnesses, use instead $d = 1 + t(p-1)$ for each integer $t \geq 1$, with $L = 10^{d-1}$. The construction above applies without change, and the d-digit interval accommodates the same p candidate endpoints. As d strictly increases, the witnesses are distinct.

For $p = 3$, take $m = 2$: $\mathcal{C}(2) = 12$, $\mathcal{L}(2) = 2$, and $3 \mid 12$ as well as $3 \mid (10^2 - 1)$. Indeed, every $m \equiv 0$ or $2 \pmod 3$ yields $\mathcal{C}(m) \equiv m(m+1)/2 \equiv 0 \pmod 3$ because $10 \equiv 1 \pmod 3$; thus there are infinitely many witnesses for 3 as well.

Finally, for $p = 2$ or 5, the first decimal digit of $\mathcal{C}(m)$ is 1. A digit rotation that moves this digit to the last place ends in 1 and hence cannot be divisible by 2 or 5. This completes the proof. □

Corollary 1 (An order-sensitive bound)**.** *Let $p \geq 7$ be prime, and write $r = \mathrm{ord}_p(10)$. There exists a witness m with*

$$m \leq 10^{2r} + pr - 1 < 10^{2r+1}.$$

Thus one may take the smaller of this bound and (4).

Proof. Set $d = 2r + 1$, $L = 10^{2r}$, $H = \mathcal{L}(L-1)$, and choose $\delta \in \{0, \ldots, r-1\}$ with $\delta \equiv -H - 1 \pmod r$. Put $m_0 = L + \delta$ and consider $m_j = m_0 + jr$ for $0 \leq j < p$. Every candidate has digit count divisible by r, since

$$\mathcal{L}(m_j) = H + d(\delta + jr + 1) \equiv H + \delta + 1 \equiv 0 \pmod r.$$

Moreover, $\delta + (p-1)r \leq pr - 1$; because $p \mid 10^r - 1$, we have $p < 10^r$, hence $pr < r10^r < 9 \cdot 10^{2r}$. Thus all candidates remain d-digit integers.

The step in Lemma 1 is now $-r/9$, not $1/9$ as in the special case $r = p - 1$. In particular,

$$\mathcal{C}(m_j) \equiv \mathcal{C}(m_0) - jr\,9^{-1} \pmod p.$$

Since $r = \mathrm{ord}_p(10) \mid p - 1$, the inverse $r^{-1} \pmod p$ exists. Choose explicitly

$$\boxed{j \equiv 9\,r^{-1}\mathcal{C}(m_0) \pmod p, \qquad 0 \leq j < p.} \tag{9}$$

Then $p \mid \mathcal{C}(m_j)$, while $r \mid \mathcal{L}(m_j)$ implies $p \mid 10^{\mathcal{L}(m_j)} - 1$. The rotation criterion (2) completes the construction. □

4 Discussion

The argument does not determine the least witness m in A181373; it proves its existence, gives an explicit (generally nonminimal) construction, and establishes infinitely many valid endpoints for each admissible prime. The only external tools are elementary modular arithmetic and Fermat's little theorem.

References

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[4] M. Ripà and E. Dalmasso, Patterns related to the Smarandache circular sequence primality problem, *Notes on Number Theory and Discrete Mathematics*, **18** (2012), no. 1, 29–48. https://nntdm.net/volume-18-2012/number-1/29-48/.

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